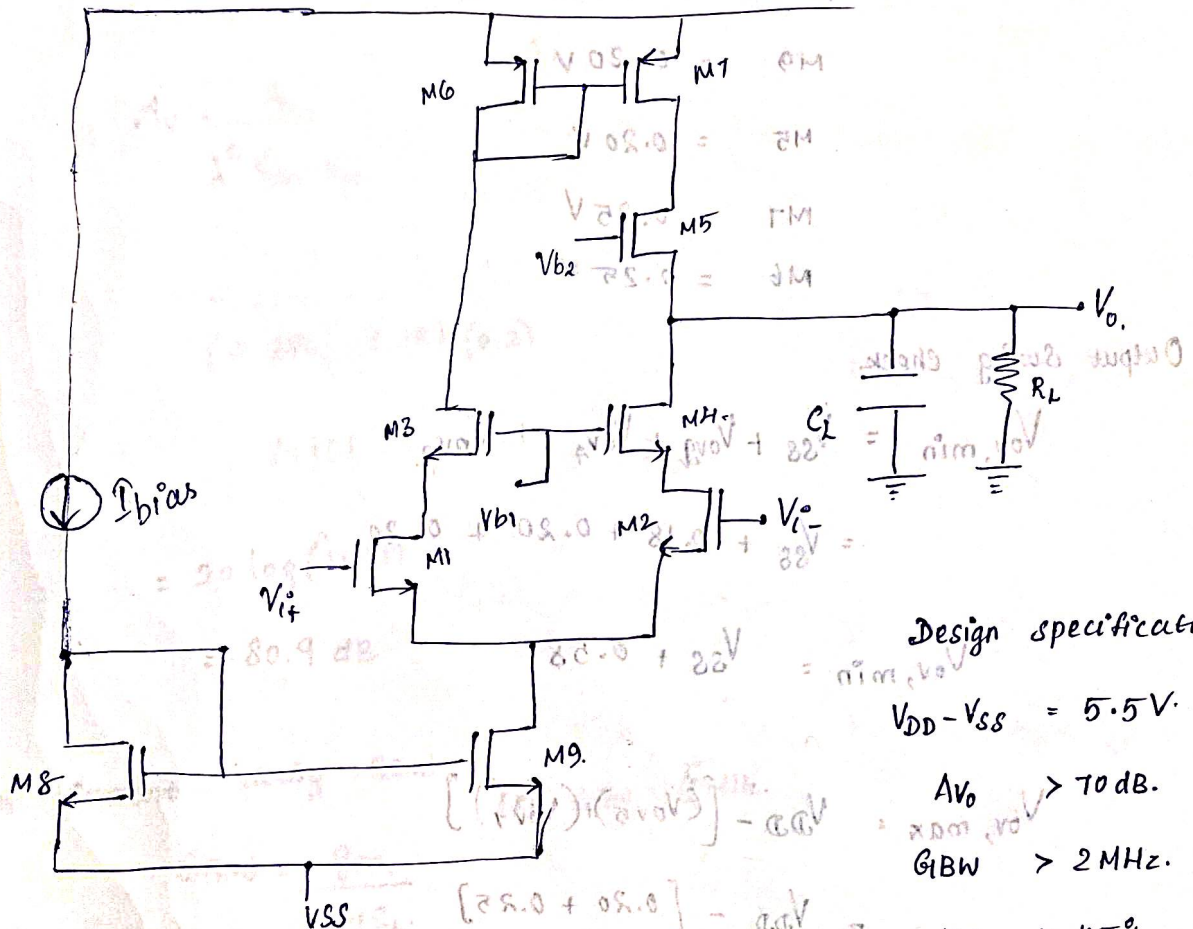


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Pre-lab - 4. : Op-Amp Design.

Design the telescopic amplifier with output dominant pole. Identify the dominant noise sources and find the expression for the input referred noise due to the dominant noise sources. Show the hand-calculations for the input referred noise and report estimated value.



Design specifications:-

$$V_{DD} - V_{SS} = 5.5V.$$

$$A_{V_0} > 70dB.$$

$$GBW > 2MHz.$$

$$PM > 45^\circ.$$

$$\text{Output swing} > 0.8V.$$

$$C_L = 2pF.$$

Considering, $W/L = 0.5/0.18$. Let $(\mu_n C_{ox}) = 167 \mu F/V$.

$$(\mu_p C_{ox}) = 47.2 \mu F/V.$$

$$\{82.0 + 24.0\} - 2.2 = 103.8$$

$$V_{B1} < 1.1V$$

choosing the Over-drive voltages,

Let the V_{ov} of input transistors,

$$M1, M2, = 0.18 \text{ V}$$

$$M3, M4 = 0.20 \text{ V}$$

$$M9 = 0.20 \text{ V}$$

$$M5 = 0.20 \text{ V}$$

$$M7 = 0.25 \text{ V}$$

$$M6 = 0.25 \text{ V}$$

$$V_{b1} = V_S + V_{ov3}.$$

$$= 0.9 + 0.2 + 0.18 + 0.20$$

$$\boxed{V_{b1} = 0.98 \text{ V}}$$

Output Swing check.

$$\begin{aligned} V_{ov, \min} &= V_{SS} + V_{ov2} + V_{ov4} + V_{ov9} \\ &= V_{SS} + 0.18 + 0.20 + 0.20. \end{aligned}$$

$$V_{ov, \min} = V_{SS} + 0.58.$$

$$V_{b2} = V_{DD} - (V_{ov7} + (V_{ov5} + V_t))$$

$$= 5.5 - (0.25 + 0.20 + 0.1)$$

$$= 5.05 \text{ V}.$$

$$\boxed{V_{b2} = 4.65}$$

$$V_{ov, \max} = V_{DD} - [(V_{ov5}) + (V_{ov7})]$$

$$= V_{DD} - [0.20 + 0.25]$$

$$= 5.5 - 0.45$$

$$= 5.05$$

$$V_{ov, \max}$$

$$V_{swing} = 5.5 - \{0.45 + 0.58\}$$

$$\boxed{V_{swing} = 4.47 \gg 0.8 \text{ V}}$$

For Gain Requirements:

Required Gain = 70 dB.

$$R_{out} \text{ for telescopic op-amp} = \frac{1}{\lambda I_D}$$

$$A_V = \frac{4}{\lambda^2 V_{ov1} V_{ov2}}$$

$$= \frac{4}{(0.18)^2 \times (0.18) \times (0.2)}$$

$$= 11111$$

$$= 20 \log(11111)$$

$$= 80.9 \text{ dB}$$

Finding Bias Current using GBW,

$$GBW = \frac{g_{m1}}{2\pi C_L}$$

$$g_{m1} = 2\pi \times 2 \text{ MHz} \times 2 \times 10^{-12} = \frac{8\pi}{\text{J}}$$

$$g_{m1} = 62.83 \mu\text{S}$$

$$I_D = g_{m1} = \frac{2I_D}{V_{ov1}} \left(\frac{W}{L} \right) \times \frac{1}{6} = 25.9 \mu\text{A}$$

$$\frac{62.8 \mu\text{S} \times 0.18}{2} = \frac{I_D}{25.9 \mu\text{A}} \Rightarrow I_D = 25.9 \mu\text{A}$$

$$I_D = 2.259 \mu A$$

$$I_{tail} = 2 \times 5.659 \mu A$$

$$I_{tail} = \cancel{4.518 \mu A} + 2 \times 6 \mu A = 12 \mu A$$

$$\text{Let } I_{tail} = 12 \mu A$$

Finding the w/L aspect ratio.

$$\left. \begin{array}{l} M1, M2 \\ M3, M4 \\ M5, M6, M7 \\ M8, M9 \end{array} \right\} \begin{array}{l} 2.259 \mu A \\ 2.259 \mu A \\ 4.518 \mu A \\ 4.518 \mu A \end{array}$$

$$\text{For } M1, M2 \quad I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_t)^2$$

$$6 \mu A = \frac{167}{2} \times \left(\frac{W}{L} \right) (0.18)^2$$

$$\frac{W}{L} \geq \text{minimum}$$

$$\frac{W}{L} = \frac{0.65}{0.18} \times 1 = 0.5 \mu / 0.18 \mu = \left(\frac{W}{L} \right)_{1,2}$$

For M3, M4,

$$6 \mu A = \frac{167}{2} \times \left(\frac{W}{L} \right) (0.20)^2$$

$$= 0.6 \times \left(\frac{0.5}{0.18} \right) = 1.1 \times \frac{0.5}{0.18}$$

$$= 0.5 \mu / 0.18 \mu = \left(\frac{W}{L} \right)_{3,4}$$

for M_5

$$\Rightarrow 6\mu = \frac{47.2}{2} \left(\frac{\omega}{L} \right) (0.20)^2$$

$$2.28 \times \left(\frac{0.5}{0.18} \right) = \frac{\omega}{L}$$

$$\left(\frac{\omega}{L} \right)_5 = (1\mu / 0.18\mu)$$

for M_6, M_7

$$6\mu = \frac{47.2}{2} \left(\frac{\omega}{L} \right) (0.25)^2$$

$$4. = \frac{\omega}{L}$$

$$1.5 \times \frac{0.5}{0.18} = \frac{\omega}{L}$$

$$\left(\frac{\omega}{L} \right)_{6,7} = 0.75\mu / 0.18\mu$$

for M_9, M_8

$$12\mu = \frac{167}{2} \times \left(\frac{\omega}{L} \right) (0.2)^2$$

$$1.3 \times \frac{0.5}{0.18\mu} = \frac{\omega}{L}$$

$$\left(\frac{\omega}{L} \right)_{9,8} = 0.65\mu / 0.18\mu$$

Satisfying the Gain Margin,

$$P_1 = \frac{1}{R_{out} C_L}$$

$$R_{out} = \frac{1}{\lambda D_D}$$

$$\lambda = 0.23$$

$$D_D = 6 \mu \cdot \left(\frac{481.0}{1} \right) = \left(\frac{2.0}{1} \right)$$

$$R_{out} = \frac{1}{0.23 \times 6 \mu}$$

$$= 0.724 \text{ Mohm.}$$

$$g_m = 62.83 \mu S \cdot \times 0.724 = 45 \text{ kHz} \quad \frac{\omega}{1} = \text{As assumed,}$$

$$P_1 = \frac{1}{\text{---}}$$

$$45 \text{ kHz} \times 2 \times 10^{-12}$$

$$= 11.11 \text{ MHz.}$$

$$\text{Phase margin} = 90^\circ - \tan^{-1} \left(\frac{11.11 \text{ MHz}}{5 \text{ MHz}} \right)$$

$$PM = 66^\circ > 45^\circ \text{ } \Leftarrow \text{Satisfied.}$$

Input - Referred Calculation of Noise,

NMOS $\rightarrow$ Input referred Noise,

$$= \frac{4kT\gamma}{g_m} \quad \gamma = \frac{2}{3}$$

$$k = 1.38 \times 10^{-23}$$

$$T = 300K$$

$$= \frac{8kT}{3 \times g_m}$$

$$\gamma = 2/3$$

$$g_{m1} = 62.83 \mu$$

$$= \frac{8 \times (300) \times (1.38 \times 10^{-23})}{3 \times 62.83 \times 10^{-6}}$$

$$V_{in}^2 = \frac{5.27 \times 10^{-16}}{3} V^2/Hz. = 1.75 \times 10^{-16}$$

$$V_{in} = \frac{0.022 \mu V/Hz}{3} \quad 1.322 \times 0.0132 \mu V$$

Finding input referred noise.

$$V_{rms} = 0.022 \mu V/Hz. \times \sqrt{GBW.}$$

$$= 0.022 \mu V/Hz \times \sqrt{5 \times 10^6}$$

$$= 49.19 \mu V. \quad 29.06 \mu V/Hz$$

(Specification met)